STAYHERE · INTERNAL ANALYSIS

Floor Plan AI Model Options

Reliability upgrade options for AI-assisted floor plan analysis, with cost projections at scale

Prepared for StayHere · October 2026

Why Reconsider the Model

StayHere's floor plan analysis feature currently runs on Google's Gemini Flash tier as its primary model, with a Groq-hosted Qwen model as fallback. Both are fast and inexpensive, but for a feature this central to listing quality, it's worth evaluating whether a more capable paid model materially improves accuracy and reliability — and understanding exactly what that would cost at real operating volume.

> **Key architectural fact:** floor plan analysis is **not a single AI call per image**. The pipeline detects rooms in one pass, then makes a separate follow-up call per detected room to find doors and windows. A typical 1–2 bedroom unit (~6 rooms) takes about **7 model calls**; a larger 3-bedroom-plus unit (~10 rooms) takes about **11 calls**. This call pattern is unchanged by whichever model answers each call — it's the same regardless of provider.

Three Model Options

1. Claude Sonnet 5.5 **$2.00 / $10.00 per million tokens (in / out)**

Best-in-class structured-output discipline. The pipeline forces every response into a strict JSON schema, which is exactly where Claude's tool-use reliability shows up. Strong spatial and document reasoning on messy, real-world floor plan photos.

2. GPT-5 **$1.25 / $10.00 per million tokens (in / out)**

OpenAI's current flagship multimodal model. Strong general vision reasoning, slightly cheaper input pricing than Claude here, and mature JSON-schema-constrained output support.

3. Gemini 2.5 Pro **$1.25–2.50 / $10–15 per million tokens (in / out)**

Same provider and SDK already integrated — effectively a model-name swap with minimal code change. Meaningfully more capable than the current Flash tier at spatial reasoning, while keeping the existing retry/fallback logic almost entirely intact.

Also worth naming as a cheaper middle ground: **Claude Haiku 4.5** ($1.00 / $5.00 per million tokens) — notably better than Flash, still far cheaper than the three options above.

The Cost Math

Each model call also carries the pipeline's existing system prompt (~450 tokens), a JSON schema (~150–200 tokens), short user text, and a representative floor plan image (~1500×1100px full image, ~900×650px per-room crops). Using each provider's official image-tokenization formula:

Model	Cost per typical unit (~6 rooms, 7 calls)	Cost per larger unit (~10 rooms, 11 calls)
Claude Sonnet 5.5	~$0.032	~$0.046
GPT-5	~$0.020	~$0.029
Gemini 2.5 Pro	~$0.019	~$0.028
Reference: Gemini 2.5 Flash (current)	~$0.005	~$0.007

Adding ~20% headroom for the occasional extra "missed rooms" pass and retries gives a realistic blended per-plan cost: **Claude ≈ $0.04**, **GPT-5 ≈ $0.03**, **Gemini 2.5 Pro ≈ $0.025**.

Daily Cost at Volume

Floor plans / day	Claude Sonnet 5.5	GPT-5	Gemini 2.5 Pro
20	$0.80	$0.60	$0.50
50	$2.00	$1.50	$1.25
100	**$4.00**	**$3.00**	**$2.50**

The Takeaway

At this volume, model choice is essentially free either way — the gap between the cheapest and priciest "world-class" option at 100 plans/day is about **$1.50/day (~$45/month)**. This isn't a cost decision, it's a reliability and accuracy decision.

Recommendation

Priority	Pick	Why
Lowest friction	Gemini 2.5 Pro	Same SDK/auth already in place — near-zero code change beyond the model config value
Second opinion / different reasoning engine	GPT-5	Genuinely different model family inside the existing fallback chain

Best structured-output correctness	Claude Sonnet 5.5	Strongest fit for ambiguous, messy real-world floor plans where schema correctness matters most

Sources: Claude pricing (claude.com/pricing) · Claude vision/image tokens (platform.claude.com/docs/en/build-with-claude/vision) · OpenAI pricing (developers.openai.com/api/docs/pricing) · OpenAI image tokenization (developers.openai.com/api/docs/guides/images-vision) · Gemini pricing (ai.google.dev/gemini-api/docs/pricing)

Analysis only — no code changes were made as part of this document.